

Combating Voice Spoofing Attacks on Wearables via Speech Movement Sequences

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Abstract—Voice assistants, increasingly integrated into wearable devices with limited human-computer interaction modalities, are susceptible to voice spoofing attacks. Such attacks exploit pre-recorded or synthesized voice commands to trick the assistants into executing actions unauthorized by legitimate users. In this work, we propose GyroTalk, a novel approach extracts individual and reliable features from speech movement sequences of users, using built-in gyroscopes in wearables, to differentiate between legitimate users and malicious attackers. GyroTalk is inspired by two critical insights. First, speech, as a highly intricate motor task, necessitates the synchronized coordination of multiple respiratory, laryngeal, lingual and mandibular muscles. These collective muscle movements propagate throughout the body, providing unique movement signatures. Second, the distinctive speech movement sequences of individual speakers, essential for generating specific words, can be grabbed by embedded IMU of wearables. We conduct a comprehensive evaluation of GyroTalk across various COTS Android devices, including smart phones, watches and glasses. Our experimental results demonstrate that GyroTalk can achieve a mean FAR of 2.23% and a FRR of 2.48%, even in the face of complicated voice spoofing attacks.

Index Terms—Voice assistant, spoofing attack, speech movement sequence, gyroscope, wearable.

I. INTRODUCTION

OVER the past decade, the prevalence of wearables has significantly increased, ranging from smart glasses, watches, and headphones to wrist-bands and jewelries. However, such wearable devices often lack adequate human-computer interfaces (i.e., limited display size and restricted input methods). As a result, traditional keyboard- or touchscreen-based interactions,

typically used in PC and mobile scenarios, respectively, are less suitable for wearables. Conversely, voice-based interactions, which utilize voice assistant apps (e.g., Google Assistant, Siri, Alexa, and Cortana) to make phone calls, send messages, and play music without typing commands, delivering a more user-friendly experience.

However, voice-based interactions are highly vulnerable to spoofing attacks due to the open sound channel, making it easy to record voice activities without being detected [1]. The widespread availability of high-quality and affordable recording/replaying devices, such as smartphones and digital voice recorders, has simplified the launch of replay attacks. Additionally, attackers can fabricate voice commands by splicing multiple voice fragments of a victim [2], [3]. Existing research indicates that, with only one-minute voice training data, attackers can build a fine-tuned model, which is capable of synthesizing voice commands highly similar to the original ones of a victim. These vivid voice commands can be, afterwards, surreptitiously injected into the sound channel [4], [5], [6], [7]. Such attacks, which can lead to serious consequences such as making fraudulent phone calls, unlocking devices and authorizing online payments, have raised significant concerns.

In academic research, liveness detection is employed to combat voice spoofing attacks by distinguishing authentic human voice samples from machine-generated ones. Current liveness detection methods are categorized into three types. First, Feng et al. utilize a high-sampling-rate (11 kHz) vibration sensor to detect vibrations from vocal cords and match them with the corresponding microphone-captured speech signals [8]. However, the sensor is regraded as a hardware extension for wearable device and needs to be placed near the throat of speaker. Second, use a camera to capture lip movements [9] or extract reliable facial features [10]. Nevertheless, such schemes are subject to lighting conditions and affordable wearable products often lack cameras. Third, ultrasonic-based approaches leverage the built-in speaker of a smartphone to emit high-frequency sound wave, then detect Doppler shift in the echo signal while a user utters voice commands, which are caused by articulatory gestures [11], [12], [13]. But the user has to hold the smartphone right in front of his/her face within an extremely short distance (2–4 cm), making it impractical for wearable devices like glasses, necklaces, and headphones.

In this work, we introduce GyroTalk, a novel liveness detection method that utilizes *Speech Movement Sequences* (SMSs). The key concept of GyroTalk is to leverage the embedded

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gyroscopes in wearables to record individual SMSs when a speaker utters voice commands. The recorded sequences are then used to verify user authenticity. GyroTalk is driven by two fundamental observations. First, voice is a task-oriented motor behavior that necessitates coordinated efforts from multiple joints and muscles. Consequently, functional movements of the jaw and head-neck regions arise from the coordinated activation of muscles in both the jaw and neck, resulting in joint motions of temporal mandibular, atlantooccipital joint and cervical spine. The synchronized movements of jaw and head-neck parts can also propagate to other distant body segments, facilitated by the interlinked human skeleton and interconnected muscle fibers. Second, even when speaking the same vocal command, individuals exhibit diversity in SMSs due to variations in muscle size, shape, and strength, as well as differences in habitual patterns of muscle force exertion on bones and joints. Such unique features can be used to catch a malicious attacker who attempts to mimic the SMS of a legitimate user.

GyroTalk provides offline user enrollment, which allows a user to repeatedly utter preselected voice commands, then records the corresponding movement sequences as training samples. In online authentication, a user only needs to utter the pre-registered voice commands once, then the system will compare the corresponding movement sequences with voice samples. However, GyroTalk currently faces three main challenges. First of all, accurately perceiving tiny movements induced by speech is not an easy task. By analyzing the perceptual data from common embedded motion sensors, we find that speech movements usually manifests itself as slight accelerations, but exhibits relatively large changes in rotation. Therefore, we leverage a gyroscope to capture SMSs.

Second, it is hard to choose movement features that precisely represent the characteristics of a speaker. Despite the widespread adoption of statistical features, which are often insufficient in accurately representing the comprehensive trend of the entire movement. In addition, directly feeding movement sequences into the classifier may lead to higher computational overhead and potential inaccuracy. To solve this problem, we choose to analyze the similarity between registered samples and test samples. To enhance the robustness of feature vectors based on similarity, we compare the test vocal commands with top- K samples.

Finally, due to the difficulty of eliminating variances induced by different speaking rhythms, it is hard to calculate the similarity between two motion signals. There are two aspects to this problem. In instances where the same speaker utters two different passphrases, varying speaking rhythms may cause corresponding syllable movements to appear at different points in the gyroscope signals. This discrepancy often results in a low similarity mistakenly. To tackle this problem, we introduce *Speech Aligned Dynamic Time Warping* (SA-DTW), a method that adjusts gyroscope signals to align with time-synchronized microphone signals for enhanced similarity comparison. When two passphrase phrases are uttered by different speakers, it is essential to avoid erroneous high similarity due to incorrect alignment of different syllabic actions. We observe that, in the best match to the template sequences, SMSs from real users

should have small aberrations. Thus, we define *Warping Score* to measure the similarity and SA-DTW distance between SMSs.

We conduct extensive real-world experiments with 15 enrolled participants on three commercially available off-the-shelf (COTS) Android devices, including a watch, a phone, and a pair of smart VR glasses. GyroTalk is capable of accurately separating authorized users from adversaries (whether human or machine) of voice assistant systems. This innovative technology can be utilized on wearable products that are worn on different parts of the body without any complicated procedures and hardware extensions. Experimental results demonstrate that GyroTalk can achieve a low mean false acceptance rate (FAR) of 2.23% and a false rejection rate (FRR) of 2.48%, even against complicated voice spoofing attacks.

Contributions: This article makes the following contributions.

- We show that the SMSs of speaking are resulted from the coordinated movements of several muscles, which can be captured by the built-in motion sensors in wearable devices. It lays the foundation of our GyroTalk design.
- Based on the above observation, we develop GyroTalk, which extracts the user-specific features of SMSs when speaking a passphrase for liveness detection. GyroTalk relies on the speech movement control system of a speaker and requires no complex operations or additional hardware, just a gyroscope, which is commonly embedded in wearable devices.
- We evaluate GyroTalk on 3 COTS Android devices, including a pair of VR glasses, a phone and watch. We also recruit 15 volunteers to conduct extensive real-world experiments and the experimental results show that even when facing advanced voice spoofing attacks, GyroTalk can achieve a mean FAR of 2.23% and FRR of 2.48%. These findings demonstrate the robustness and easy deployment of GyroTalk.

Organization: The paper is organized as follows: Section II reviews existing methods for detecting voice spoofing attacks and their limitations. Section III introduces the GyroTalk system and attack model. Section IV details the design of GyroTalk, including data collection and the use of SA-DTW for similarity measurement. Section V presents implementation and evaluation results. Section VI discusses limitations and future research directions. Section VII concludes the paper, summarizing contributions and the effectiveness of GyroTalk in preventing voice spoofing attacks on wearables.

II. RELATED WORK

Existing research on detecting voice spoofing attacks primarily moves in two directions: liveness detection and channel characteristic analysis.

A. Liveness Detection

Liveness detection systems are intended to distinguish authentic human voice samples from machine-generated ones, and these systems can be divided into three categories.

1) *Vibration-Based Detection Methods:* High-frequency speech signals are assessed using an acceleration sensor

and compared with voice signals from a microphone. A voice attack is detected if the correlation between these signals falls below a threshold [8]. This approach requires an additional sensor placed near the throat to capture vocal cord vibrations. Wan et al. find that microphone vibrations caused by breathing produce a unique pop noise, hard for attackers to replicate [14], but this method is susceptible to ambient noise and needs the microphone close to the mouth. Liu et al. introduce a multi-modal system using mmWave vibrations and audio signals [15], facing hardware dependency and user inconvenience. Han et al. use accelerometer data for speech vibrations [16], but it is sensitive to ambient noise and requires careful calibration. Embedded motion sensors in mobile devices are used to extract, classify and verify subtle muscle tremors that occur naturally in the body during use of the device [17], or capture biometrical features when a user shaking the phone in his customized way [18].

2) *Conversation-Face Based Detection Methods*: Detection methods based on facial and lip motion analysis are used to identify the speakers. These methods calculate the similarity between recorded voice and facial information to prevent attacks [10], [19], [20], [21]. Nevertheless, the effectiveness of these schemes can be influenced by illumination. On the other hand, accessing the camera has the potential to compromise sensitive user privacy. To overcome the disadvantages of camera-based approaches, Xu et al. utilize a commercially available mmWave radar, moving along a set path, to scan the human face, and the reflected signals can capture facial biometric and structural features [22]. However, their device is required to be directly placed in front of the human face within a limited range of angles.

3) *Ultrasound-Based Detection Methods*: Upon a passphrase spoken by a user, the built-in speaker emits ultrasonic waves. The microphone then captures the Doppler shift in echo signals, reflected by various articulators of the sound system [11], [12]. The disadvantages of these methods are that the device must be placed near the mouth of a user and certain posture is required to ensure accurate capture of the reflected signal. Wang et al. employ Wiener deconvolution preprocessing and device-specific convolution to boost ultrasound-based anti-spoofing system [23]. However, it mandates collecting pulse response data from target devices and conducting data augmentation and training for each unseen device, potentially requiring extra equipment and data collection, thereby hampering its universality and quick deployment on new devices. Lee et al. propose a sonar-based detection scheme, which verifies sound source consistency by emitting an inaudible sound to identify the relative location of the user via a smart speaker [24]. Unfortunately, this method consumes too much power for wearable devices. Jiang et al. present a software-based anti-spoofing approach for voice authentication on smartphones. It leverages unique sequences to describe the relationship between pop noises and phonemes, then calculates the similarity between estimated and actual pressure signals [25]. However, their scheme requires the smartphone right in front of the mouth of speaker within a short

distance (i.e., 2-4 cm), in order to detect the ultrasonic signal effectively.

B. Channel Characteristic Detection

Channel characteristic-based schemes detect differences in data between Hi-Fi recordings and original sound channels. These schemes use the Universal Background Model (UBM) to model the channel, and analyze the silent segments of voice data to verify if the authenticated channel matches the training voice channel [26]. While these methods provide valuable insights into detecting voice attacks, it is important to note that due to the lower amplitude and susceptibility to noise pollution of the silent segments, additional measures are required for comprehensive protection. In recent years, Meng et al. propose a multi-channel audio-based scheme which leverages the layout characteristics of microphone arrays in smart speakers and the channel properties of sound propagation to achieve the discrimination between genuine human and replayed speech [27], [28]. Yang et al. leverage differences between multiple microphone channels to extract acoustic dynamics features, distinguishing between genuine voice commands and spoofing commands played through speakers [29]. However, our GyroTalk is primarily intended for wearable devices, which typically do not feature such microphone arrays. Moreover, human breath will leave a trail in the WiFi Channel State Information (CSI) during the course of speaking commands, PRADHAN et al. use changes in the CSI traces to detect breath to verify human presence [30], but this approach is restricted to scenarios where WiFi is present.

Overall, our GyroTalk is a new form of liveness detection method for wearable devices, which can be worn in flexible ways (e.g., hand-held, wrist-, and head-worn). Users have no learning curve to adapt to the system. Meanwhile, GyroTalk does not require additional hardware extension, and there is no risk of leaking user privacy.

III. PRELIMINARIES

A. System and Attack Model

GyroTalk aims to ensure the safety of speaker recognition tasks, that is, text-prompted voice authentication. In enrollment, more specifically, a user is requested to utter a prompted phrase predefined by the provider several times. In future verification, the user attempts to get a pass by speaking the enrolled phrase. GyroTalk can be deployed on a variety of COTS wearables such as necklaces, glasses, bracelets, helmets, and mobile phones. These devices must be equipped with gyroscopes and have computing capabilities. Additionally, they should be carried or worn by users to allow the inertial sensor to detect micro-movements induced by speaking.

Suppose an attacker who seeks to deceive the voice authentication system of the victim by using inverteous passphrases. The attacker has two options to carry out spoofing attack: a replay attack (i.e., recording the victim's voice beforehand and playing it back during authentication) or a voice impersonation attack (i.e., injecting a modulated or imitative vocal command

into the audio channel). Furthermore, we assume that a malicious attacker can obtain (i.e., have physical access to) the victim's device and master the detail of GyroTalk. Specifically, we consider two attack scenarios as follows:

- **Scenario-A: Spoofing Attack without Prior Knowledge.** In this scenario, the attacker wears the victim's device and silently mouthing the target vocal command while replaying or broadcasting it without making audible sound. Besides, the attacker has no knowledge of how victim's body moves during passphrase articulation.
- **Scenario-B: Spoofing Attack with Prior Knowledge.** This attack scenario follows the same procedure as a spoofing attack without prior knowledge, but with an additional step of the attacker first observing how the victim delivers a voice command, i.e., filming a clip of the victim and rehearsing before executing the attack.

B. Speech Movement Sequence

Typically, the activity of speaking involves the utilization of air pressure generated by the lungs to produce sound through the laryngeal valve. Subsequently, the sound is transformed into various consonants and vowels as it is modified by the vocal tract. Generating specific syllables involves the recruitment of a set of predominantly unvarying muscles, which triggers synchronized gradual motions of diverse body regions, spanning from the diaphragm to the lips [31].

Definition 1: A *Speech Movement Sequence (SMS)* is defined as a sequence of intricate body motions that occur during speaking, which involves:

- **Immediate Synchronous Motions:** the simultaneous contraction of more than 100 muscles during the vocalization process. These muscles include the septum, pectoralis major, mandibular, buccal, and pectoral muscles, which work together on the different vocal organs (e.g., lungs, throat, tongue, lips, etc.) to produce a precise sound. The coordination of these muscles is an unconscious response of the autonomic nervous system and does not require conscious control by the brain. When we speak, these muscles work in coordination through rapid signaling. In order to produce the desired sound, the brain sends signals to the muscles and tells them which contractions and relaxations to perform.
- **Traction Induced Cascades:** the interactions and mechanical properties of tissues. When a muscle group produces motion through traction, it can cause remote muscle groups and body parts to move and shift due to the interconnection of muscle fibers and attachment to the skeletal system. In the human skeletal systems, there exist complex mutual interactions among muscles, bones, and other soft tissues. Each muscle is composed of multiple muscle fibers that are interconnected by inherent proteins. Meanwhile, the muscle tissue fits snugly against the bone and is anchored to the bone by tendons (cord-like substances of connective tissue). Therefore, when a muscle group shortens or stretches, it applies force and stretches adjacent muscle and bone tissue.

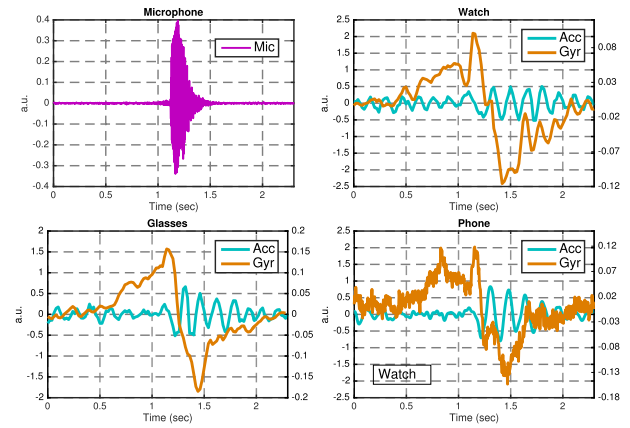


Fig. 1. When a speaker pronounces "home", the microphone picks up the raw audio data; meanwhile, the accelerometer and gyroscopes of smartwatches, glasses and smartphones record the corresponding motion sequences.

C. Capturing Diversity of SMS

We hypothesize that the contraction of specific muscle groups and body segment motions involved in voice production can be captured using motion sensors. Therefore, to capture the body motions of a speaker, we utilize both the accelerometer and gyroscope sensors commonly furnished in wearable products, which allows us to accurately track their physical movements. Based on our research, we determine that the gyroscope is a more suitable sensor for capturing speech movements due to its higher sensitivity than accelerometer. This insight indicates the gyroscope is a better choice for tasks that require precise measurement of motions during speaking, which may be due to the fact that speech movements tend to involve relatively small accelerations and larger rotations of body segments. Through detecting the output signals from a gyroscope, then conducting preprocessing and calculation, we can derive the moving direction and attitude information of a user. Fig. 1 demonstrates the original audio data recorded from a microphone positioned at the top left corner while a speaker utters the word "home". Additionally, it exhibits movement sequences collected from accelerometers and gyroscopes of different wearable devices, including a pair of VR glasses, a smart watch, and phone. These movement sequences are corresponding to the uttered word and recorded with the audio data simultaneously. The figure demonstrates that speech movements in gyroscope data can generate more significant fluctuations, even when measuring remote body parts far from the vocal tract (i.e., hand and wrist).

In addition, we believe that, when a person utters a specific voice command, the muscles of larynx and mouth will move in concert to produce a series of SMSs, which exhibit spatial-temporal characteristics that are not only unique among each individual, but also relatively consistent when the same person speaks. Specifically, this is a result of individual differences in the size, shape, and strength of muscles, bones, and joints, as well as variations in the way muscles exert force in different individuals. The uniqueness and consistency of these movement sequences also mean that they can be regarded as biometric features to identify legitimate users. Fig. 2 displays the time

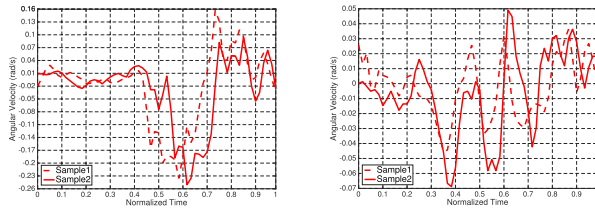


Fig. 2. Gyroscope signals of a smart watch sampled during volunteer *A* (left) and *B* (right) pronouncing “America” twice.

series of gyroscope data from a smart watch recorded among two volunteers, *A* (left) and *B* (right), each pronounces “America” twice. The result demonstrates high intra-sample similarity from the same speaker, but significant differences between samples from different speakers.

IV. DESIGN OF GYROTalk

A. Overview

The core idea of GyroTalk is to utilize the built-in gyroscopes of COTS wearable products to capture and identify unique and robust features in the SMSs of users. When a speaker utters a vocal command into a wearable device, both the voice and body movements are recorded by its microphone and gyroscope, respectively. After passing voice authentication via the microphone signal, the corresponding gyroscope data will undergo further verification by GyroTalk.

Fig. 3 shows the overall architecture of GyroTalk. The GyroTalk authentication process includes two stages: offline training and online authentication. During offline training, a user utters a vocal command multiple times to capture voice and movement signal templates. These templates are then utilized for classifier construction. Subsequently, in online authentication, test inputs will be fed into the well-trained classifier, and then compared with the established templates to verify whether it originates from a registered user.

There are two reasons why original samples are unsuitable for classifier training. First, the length of samples may differ due to variable pronouncing speed among speakers. Second, samples in GyroTalk may contain a substantial amount of data points, particularly for longer vocal commands. This often requires numerous training samples to achieve optimal accuracy, which could place an additional registration workload on users. An optimal solution is to derive feature vectors by extracting effective features from original signals. However, this approach is non-trivial. The conventional statistical features commonly employed in time series analysis, such as mean, standard deviation, and extreme values, are insufficient for classification purposes because the trend in the time series of gyroscope data, which is crucial for classification, cannot be effectively captured using these features. Stated differently, features in a data sequence are expected to appear at specific positions. As an illustration, Fig. 4 depicts two segments of gyroscope data collected from glasses worn by different speakers uttering “do”. Despite their similar maximum and minimum values and standard deviations, the

TABLE I
COMMAND LIST

Command		
2-3 words (<i>nine</i> commands)		
1. Take a photo.	2. Turn off camera.	3. Call Emily.
4. Open WeChat.	5. Play some music.	6. Lock phone.
7. Turn off Wi-Fi.	8. Check my voice-mail.	9. Next song.
4-5 words (<i>ten</i> commands)		
10. Navigate to my home.	11. Show me my first message.	
12. Turn off all alarms.	13. Show me the nearest mall.	
14. Send an email to John.	15. Go to google scholar.	
16. What’s my next appointment?	17. Do you speak Morse code?	
18. What is the weather like?	19. What’s 299 divided by 8?	
6-7 words (<i>eleven</i> commands)		
20. What is the meaning of life?	21. What song am I listening to?	
22. What is my schedule for today?	23. Set an alarm for 6 pm.	
24. Did the Golden State Warriors win today?	25. What are some attractions in City?	
26. What time is it now in Shanghai?	27. Show me pictures of Mount Rushmore.	
28. Show me restaurants near my campus.	29. Who is the Producer of Star Wars?	
30. Do I need an umbrella today?		

two segments exhibit different shapes, emphasizing that relying solely on statistical features is not sufficient.

In this paper, we present the Speech Aligned DTW (SA-DTW) algorithm, which facilitates the comparison of shape between two movement sequences that are aligned with speech. When an individual gives two vocal commands, the resulting movement sequences will carry a small SA-DTW distance and a short warping path. SA-DTW is utilized to extract distance vectors from SMSs pairs. During the classifier training process, we utilize both SA-DTW distance and warping score.

B. Data Collection

We develop an Android application that can simultaneously collect microphone and gyroscope signals with sampling rates of 44.1 kHz and 50 Hz, respectively. The experimental setup during the data collection is shown in Fig. 5. We deploy and test the application on three devices: a smartphone (Google Nexus 5X), a smart watch (MOTO 360), and a pair of VR glasses (Storm Mirror). 15 volunteers are recruited for the study, including 4 women and 11 men aged between 21 and 27 years old. During the data collection, each participant wears the watch on their non-dominant wrist, holds the smartphone in their dominant hand, and wears the glasses on their head. When giving a voice command, we instruct the volunteers to naturally bend their forearm with the smart device and hold it in front of their chest. It’s noteworthy that the participants are allowed to take a break or quit at any time.

As a preparation, each volunteer preselects 10 commands from Table I, which categorizes 30 voice commands into three groups based on their length, ensuring an approximately equal number of commands chosen from each group. Besides, volunteers are allowed to negotiate, aiming to have each command

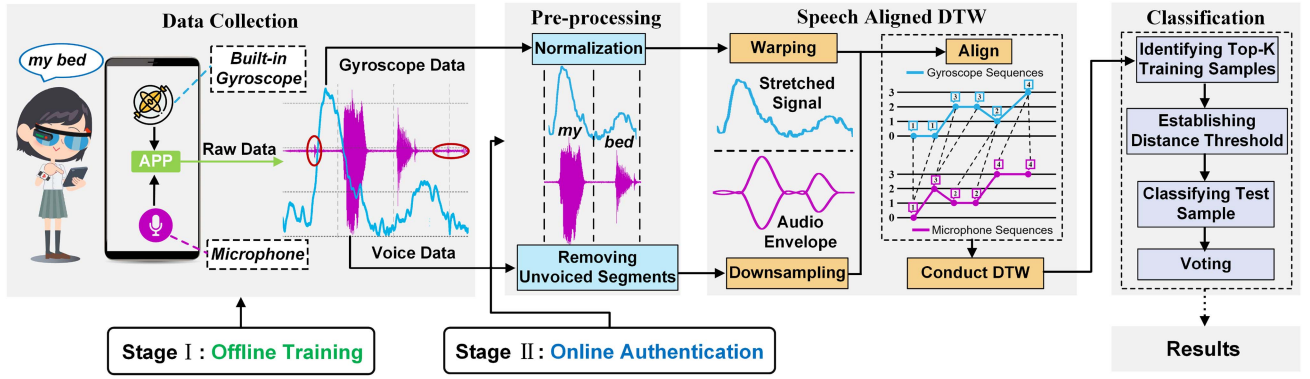


Fig. 3. The overall architecture of GyroTalk.

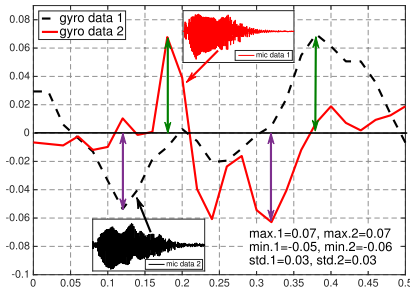


Fig. 4. A comparison of two segments of gyroscope data from different speakers, with different shapes but similar *maximums*, *minimums*, and *standard deviations*.

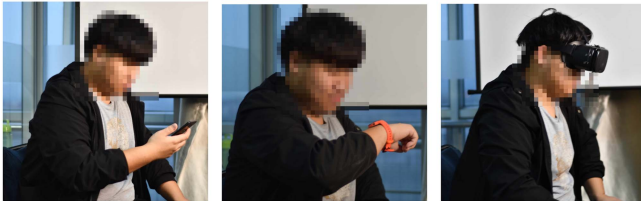


Fig. 5. Data Collection Experimental Setup for GyroTalk. smartphone (left), smartwatch (middle) and VR glasses (right).

selected 4 to 6 times, with an expected average of 5. Then, we collect data in three traces as follows:

- **Trace-A: One-month Data Collection.** During this period, each volunteer speaks 10 times per day for each selected command in various speaking volumes and paces. Specifically, every command is uttered 10 times in a standard speaking tone with 2 times each at regular, slow, and fast pace. In addition, each command is spoken at a higher and a lower volume 2 while retaining the standard speaking pace. To examine the potential impact of body postures on GyroTalk, we instruct volunteers to conduct data collection while assuming three different postures (i.e., *standing*, *sitting straight*, and *hump*) over the course of the initial, middle, and final 10 days, respectively. For every command, we gather 300 microphone signals and corresponding gyroscope data per volunteer. Additionally,

we record the data collection process with a video camera for potential use in impersonation attacks (i.e., spoofing attacks with priori knowledge).

- **Trace-B: Spoofing Attacks.** 5 participants are selected to act as victims and the remaining 10 as attackers. First, each attacker launches spoofing attack without priori knowledge (see Scenario-A) by replaying each command of victims 10 times. In the next scenario (Scenario-B), attackers watch the videos showing how victims perform during speaking voice commands. They then proceed to repeat each victim's command 10 times, making an effort to mimic their speaking styles. Therefore, for both types of spoofing attacks, each victim's command is subjected to 100 attack attempts.
- **Trace-C: One-week Data Collection after Six Months of Trace-A.** Each volunteer utters every selected command 10 times daily, maintaining a regular speaking volume and pace.

C. Pre-Processing

Assume we collect a pair of time-synchronized sequences from a microphone and gyroscope. We conduct a two-step preprocessing as follows:

- **Step-1: Removing Unvoiced Segments.** Since we only need to analyze the movements with human speech, we utilize *silence removal* [32] to remove the unvoiced segments in the audio signal. Specifically, given an audio signal V , it can be cut into multiple segments according to the voice commands, along with the corresponding gyroscope signal G , resulting in several pairs of microphone and gyroscope data segments. Each of these segments is represented as $S_i = (V_i, G_i), i \geq 1$, where V_i and G_i denote the audio and movement segments, respectively.
- **Step-2: Normalization.** When users speak, the volume of their voice and the intensity of their motion while speaking affect the amplitude of the gyroscope signal. We normalize the amplitude of each G_i segment of the gyroscope signal to remove the impact of speech volume on the signal, which aims to ensure the amplitude of the gyroscope signal will

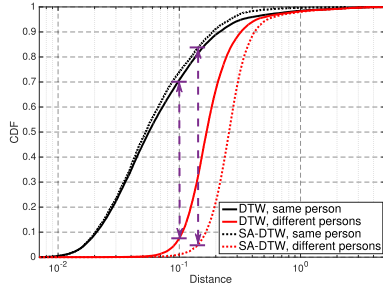


Fig. 6. DTW and SA-DTW distances between gyroscope signals coming from different and same persons shown in CDF curves.

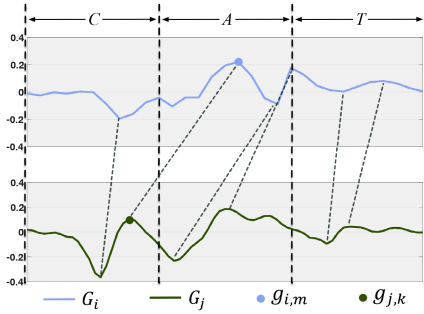


Fig. 7. An example of a mismatch. Different segments of the gyroscope signals are mistakenly matched.

not be affected by changes in the intensity or volume of the speech movements.

D. Speech Aligned DTW Based Similarity

The Dynamic Time Warping (DTW) algorithm is commonly used to compute the distance between two segments of gyroscope signals with different lengths. However, the way of using only the DTW distance to determine whether two gyroscope signal segments belong to the same person might lack sufficient accuracy. In Fig. 6, the solid line depicts the Cumulative Distribution Function (CDF) curves of the DTW distance between two gyroscope signal segments corresponding to the same voice command from the same speaker and different speakers. We can notice that narrow DTW distances also occur between gyroscope signals belonging to different speakers, e.g., 30% of the signals have distances less than 0.15. In the case where gyroscope signals are blindly warped, they could lead to segments matching up that are part of different phonemes, which would result in a short DTW distance, but a longer warping path by chance. Such mismatching is illustrated in Fig. 7, where the data point $g_{i,m}$ belonging to 'C' is incorrectly matched to $g_{j,k}$ belonging to 'A'. Hence, it is crucial to align gyroscope signals to their corresponding audio signals before comparison. In addition, it is essential for gyroscope signals from the same speaker to exhibit minimal distortion on optimal matching, as well as a tiny warping distance.

To eliminate such mismatching during signal comparison, we introduce the SA-DTW distance. We perform DTW on the microphone signals of two segments S_i and S_j , and record the corresponding warping function. To acquire the voice-aligned

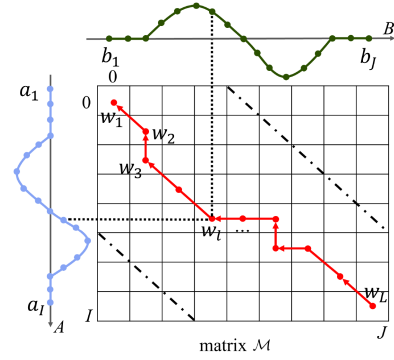


Fig. 8. An example of the resulting accumulated cost matrix $\mathcal{M}$ and its corresponding warping path $\mathcal{W}$.

gyroscope signals, we apply this warping function to the corresponding gyroscope signals. Next, we determine the SA-DTW distance by calculating the DTW distance between them, along with the associated warping score. This information is then combined into a two-tuple distance representation, capturing the similarity between the gyroscope signals. Fig. 9(a) and (b) show the joint distribution functions of DTW with the corresponding warping score, and SA-DTW with the corresponding warping score, for two gyroscope data segments from the same person and different persons, respectively. It can be seen that the two-tuple of distance is more effective for distinguishing speakers. These two-tuple distances are utilized as inputs for the subsequent classifier. Specifically, the following steps need to be executed.

1) *Downsampling Audio Signal*: After analyzing the signal sampling rates, we discover that the gyroscope signals cannot be directly subjected to the warping function between V_i and V_j due to the considerably smaller sampling rate in comparison to the microphone signals. To address this issue, we downsample the microphone signals before conducting DTW. Downsampling is commonly accomplished by decimating the microphone signal with an integer factor D such that only the D -th sample is kept. However, since the sampling rate of microphone is 882 (i.e., 44100/50) times higher than that of gyroscope, this significant disparity in sampling rates may result in the omission of peak values from the original signals, rendering the resampled signal notably dissimilar to the original. To avoid this, we choose to extract the envelope of V_i and then extract the upper part of this envelope at a sampling rate of $D=882$, resulting in a downsampled audio sequence. The determination of the envelope is performed by spline interpolation between local maxima, and the sample interval between these local maxima is at least σ (σ is set as 1,500 empirically).

2) *Warping Gyroscope Signal*: Given the downsampled signals of V_i and V_j , denoted by $A = \{a_1, \dots, a_I\}$ and $B = \{b_1, \dots, b_J\}$, we conduct DTW on them. To align A and B with DTW, we construct an $I \times J$ accumulated cost matrix $\mathcal{M}$, where the (i, j) th element in $\mathcal{M}$ is defined as $d(i, j) = (a_i - b_j)^2$, implying the alignment of a_i and b_j . During calculation of the DTW distance between A and B , we obtain a corresponding warping path $\mathcal{W} = w_1, \dots, w_L$ simultaneously, (where $w_l = (i, j)_l$, and $\max(I, J) \leq L \leq I + J - 1$),

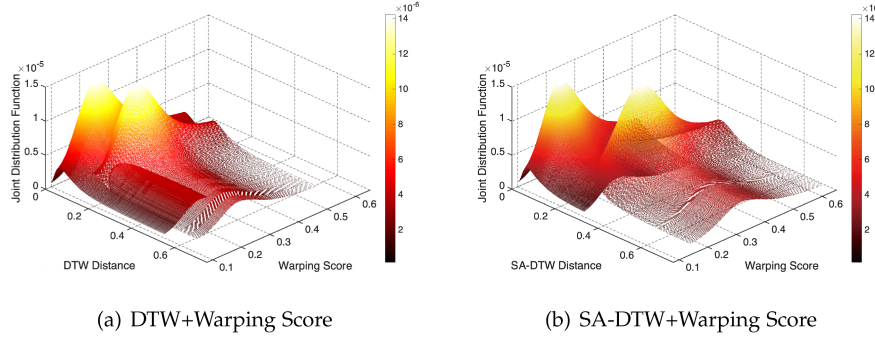


Fig. 9. The joint distribution functions of SA-DTW and DTW of two separate gyroscope data segments from the same and different persons, along with their corresponding warping scores.

on which $\frac{1}{L} \sqrt{\sum_{l=1}^L d(w_l)}$ reaches its minimum. It implies that the shortest warping path coincides with the diagonal line $j = i$ of the matrix when A equals to B , and long path means severe warping. Fig. 8 shows an example of resulting matrix $\mathcal{M}$ and the purple line indicates the corresponding warping path. Then, we stretch $G_A = \{g_1^{(A)}, \dots, g_I^{(A)}\}$ and $G_B = \{g_1^{(B)}, \dots, g_J^{(B)}\}$ according to $\mathcal{W}$, and obtain the stretched gyroscope signals $G'_A = \{g_{(i_1)}^{(A)}, \dots, g_{(i_L)}^{(A)}, \dots, g_{(i_L)}^{(A)}\}$ and $G'_B = \{g_{(j_1)}^{(B)}, \dots, g_{(j_L)}^{(B)}, \dots, g_{(j_L)}^{(B)}\}$, where i_l and j_l refer to the value of i and j in w_l . Notice that the lengths of G'_A and G'_B equal to that of A and B , respectively. For example, suppose the optimal warping path $\mathcal{W} = \{(0, 0), (1, 1), (2, 1), (3, 2), (4, 3)\}$, and given the original gyroscope signals

$$G_i = \{g_1^{(i)}, g_2^{(i)}, g_3^{(i)}, g_4^{(i)}, g_5^{(i)}\}$$

$$G_j = \{g_1^{(j)}, g_2^{(j)}, g_3^{(j)}, g_4^{(j)}\},$$

the stretched gyroscope signals G'_i and G'_j are

$$G'_i = \{g_1^{(i)}, g_2^{(i)}, g_3^{(i)}, g_4^{(i)}, g_5^{(i)}\}$$

$$G'_j = \{g_1^{(j)}, g_2^{(j)}, g_2^{(j)}, g_3^{(j)}, g_4^{(j)}\}.$$

3) *Measuring VA-DTW Distance and Warping Score:* After getting G'_i and G'_j , we calculate the DTW distance between them, denoted by $\mathbb{C}(G'_i, G'_j)$, which is actually the VA-DTW distance of G_i and G_j .

Definition 2. Given two sequences A and B with same length, i.e., $A = a_1, a_2, \dots, a_i, \dots, a_I$, $B = b_1, b_2, \dots, b_j, \dots, b_J$, where $I = J$. The area enclosed with the diagonal line $j = i$ of the matrix $\mathcal{M}$ and the warping path $\mathcal{W}$, denoted by $\mathbb{R}(\mathcal{W})$, can be calculated as

$$\mathbb{R}(\mathcal{W}) = \sum_{l=1}^L \left| \frac{(i_{l+1} - i_l)[j_l + j_{l+1} - \frac{J}{I}(i_l + i_{l+1} - 2) - 2]}{2} \right|.$$

Then the *Warping Score* of A to B is

$$\Phi(A, B) = \frac{2\mathbb{R}\mathcal{W}}{I^2}.$$

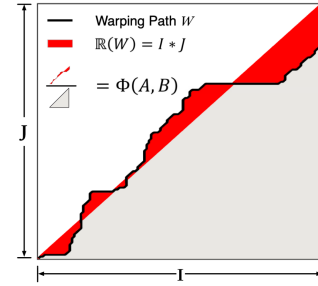


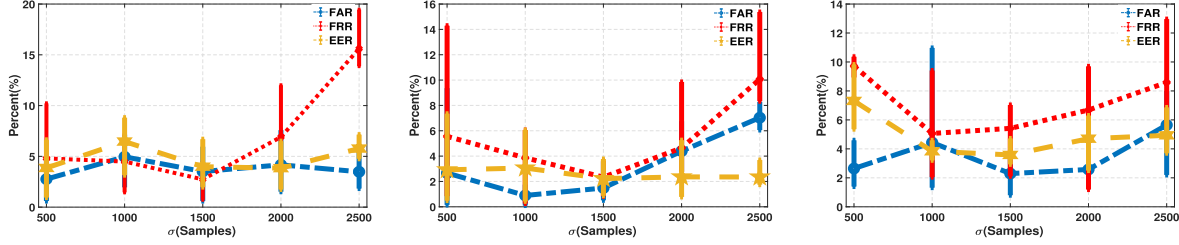
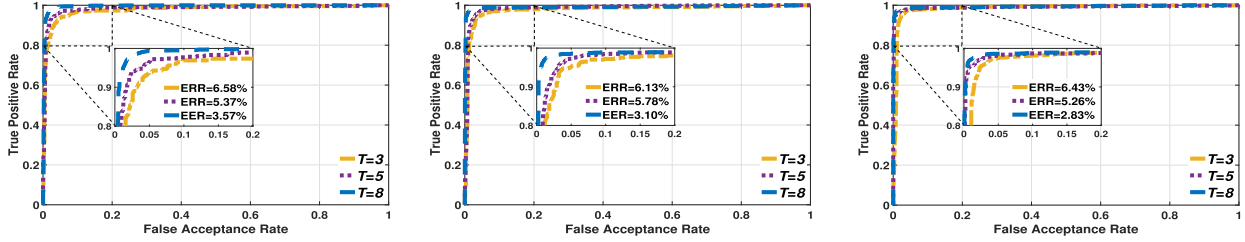
Fig. 10. An example of calculating the warping score of A to B .

An example is illustrated in Fig. 10, and the value of $\Phi(A, B)$ is between 0 to 1. Now, we compute the warping score of G'_i to G'_j , i.e., $\Phi(G'_i, G'_j)$, and form a tuple of distance, i.e., $(\mathbb{C}(G'_i, G'_j), \Phi(G'_i, G'_j))$.

E. Classification

In the offline training stage, each collected training template, comprising a pair of voice and movement signals, is subjected to mutual comparison. Specifically, we compare two sets of microphone and gyroscope data signals to each other, denoted as S_i and S_j . First, we segment them into two signal sequences, $S_{i_1}, \dots, S_{i_n}$ and $S_{j_1}, \dots, S_{j_n}$, according to Section IV-C. Then, we construct a distance vector $D_{i,j}$ between them, which consists of elements such as $\mathbb{D}(G'_{i_1}, G'_{j_1}), \Psi(G'_{i_1}, G'_{j_1}), \dots, \mathbb{D}(G'_{i_n}, G'_{j_n}), \Psi(G'_{i_n}, G'_{j_n})$. We input all distance vectors, obtained by comparing training samples, to a one-class classifier, such as KNN or SVM.

Furthermore, to enhance the efficiency and robustness of online authentication, we introduce a step to identify the top- K training samples from a given set of $\mathbb{N}$ samples. These top- K samples represent the centroid of the sample space and are best suited to represent the training samples. To identify them, we first calculate the SA-DTW distance between each sample and all training samples. Then, we choose the K samples that have the smallest average distances to other samples. By focusing on these top- K samples instead of all samples during online authentication, we can reduce calculation burden and increase

Fig. 11. Error rates with different σ (left: watch, center: glasses, right: phone).Fig. 12. ROC curve with different T (left: watch, center: glasses, right: phone).

robustness against noise in training samples. Overall, this approach provides a qualitative assessment of the sample space and improves the effectiveness of online authentication.

Online authentication relies on the evaluation of distance vectors to determine whether a test sample S_i should be accepted or rejected. This is achieved by comparing S_i with the top- K samples, which results in K distance vectors $D_{i,k}$ ($1 \leq k \leq K$). The center of these vectors, denoted as $\bar{D}_i$ is then computed by taking their average, i.e., $\bar{D}_i = \frac{1}{K} \sum_{k=1}^K D_{i,k}$. Finally, the classifier evaluates $\bar{D}_i$ to make a judgment on whether to accept or reject S_i .

V. EVALUATION

We carry out a comprehensive evaluation of GyroTalk using the traces described in Section IV-B. The data from both the gyroscope and microphone are sent to a PC for later analysis.

For the majority of experiments, we use *Trace-A*, except for spoofing attacks and long-term performance tests (as described in Sections V-C and V-B), where we utilize *Trace-B* and *Trace-C* correspondingly. To prove the efficacy of SA-DTW, we perform a comparative analysis with two comparable methods. Specifically, these methods involve generating distance vectors between the two gyroscope signals using traditional DTW distance and the DTW distance plus the corresponding warping score, respectively. We also examine the corresponding warping score for each method in comparison to GyroTalk. In order to ensure the robustness and applicability of our study, the postures we choose are the most common ones for using smartphones and wearable devices in our daily life. Moreover, collecting data of multiple poses for training can bring diversity, which helps to analyze the influence of different postures (i.e., body mechanical structures) on the performance of GyroTalk.

Metrics: We use the FAR and FRR to evaluate GyroTalk. The FAR measures the proportion of illegitimate testing inputs that are falsely accepted, while the FRR calculates the percentage of valid testing inputs that are incorrectly rejected. To assess the diagnostic performance of a binary classifier, we vary its discrimination threshold using the Receiver Operating Characteristic (ROC) curve. The Equal Error Rate (EER) is calculated at the intersection of the FAR and FRR curves on the ROC curve.

A. Micro-Benchmarks

Within this subsection, we evaluate how GyroTalk performs when subjected to various real-world factors.

Efficacy of σ : In Fig. 11, the mean, maximum, and minimum values of FAR, FRR, and EER across all participants are depicted as a function of σ on various devices. This experiment involves three-word commands, with σ ranging from 500 to 2,500 at intervals of 500. A one-class SVM classifier is used, with K set to 4. The corresponding analysis of the resulting envelope shows that too small a value of σ causes the envelope to contain unnecessary detail, while too large a value of σ distorts the envelope, leading to significant deviations between the downsampled signal and the original signal. Hence, σ is set to 1500 for the subsequent experiments.

Efficacy of T : We evaluate the performance of GyroTalk across various training data size, specifically at $T = 3, 5$, and 8 , with K fixed at 4. We use a SVM classifier (one-class), which discrimination threshold is increased from 0.0 to 1.0 at a 0.01 interval and its corresponding ROC curves are shown in Fig. 12. Our analysis reveals that increasing T results in a gradual drop in the average EERs across all three devices studied. Notably, GyroTalk achieves the best performance when $T = 8$. It is essential to note that the computational burden of the online authentication stage remains unaffected by the value of

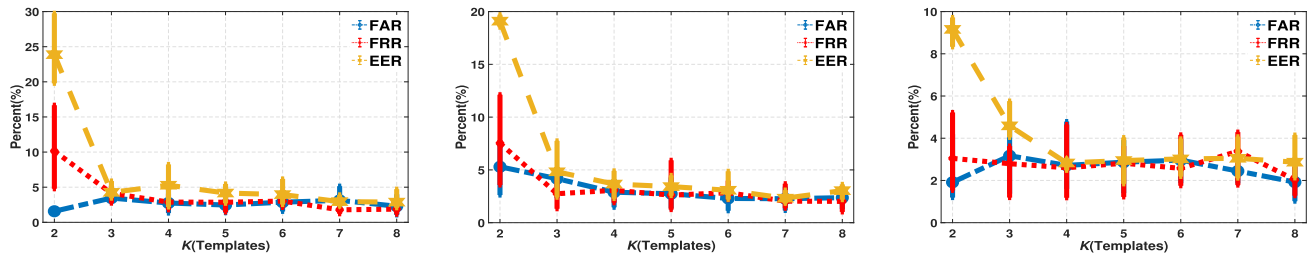
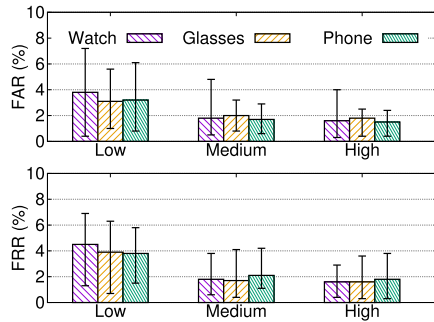
Fig. 13. Error rates with different K (left: watch, center: glasses, right: phone).

Fig. 14. FAR and FRR with different speaking volume.

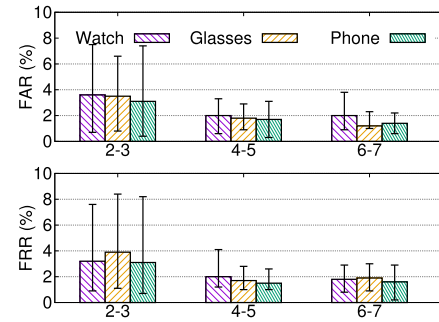


Fig. 16. Long term performance.

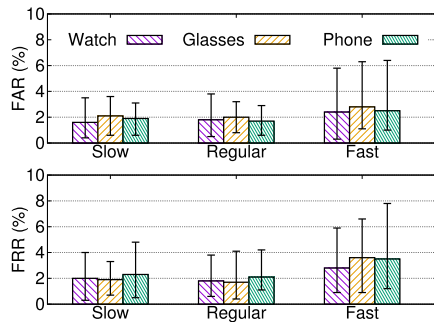


Fig. 15. FAR and FRR with different speaking pace.

T , as only the Top- K samples are compared against the testing samples. Thus, we utilize a training sample size of *eight* for the remainder of this study.

Efficacy of K : In Fig. 13, the mean, maximum, and minimum FAR, FRR, and EER for all volunteers are plotted against K on three different devices. The experiment involves commands with three words, with K varying from 2 to 8 at an interval of 1. The results show that all three error rates decrease significantly when K is increased from 2 to 4 and gradually as K is further increased to 8. Additionally, even with a small value of K , such as 3, high performance is achieved across all terminals. In the process of online authentication, the top- K samples are used as reference to compare with the testing sample. Large K value can result in significant computational overhead and long authentication latency. Therefore, for the rest of the study, K is set to 3 as

TABLE II
COMPARATIVE ANALYSIS OF TWO CLASSIFIERS

Classifier	Watch		Smartphone		Glasses	
	FAR	FRR	FAR	FRR	FAR	FRR
KNN	4.09%	5.85%	5.03%	5.17%	3.82%	4.79%
SVM	3.45%	4.13%	3.24%	3.53%	3.01%	3.35%

it delivers the optimal trade-off between computation cost and performance.

Influence of Classifier: In this study, we evaluate the performance of two one-class classifiers: K-Nearest Neighbor (KNN) [33] and Support Vector Machine (SVM) [34]. We conduct multiple tests with KNN, varying the parameter κ from 1 to 10, and determine that the optimal value is 3. For SVM, we adopt Radial Basis Function (RBF) as the kernel function, and perform a grid search with cross-validation on the training set to determine the values of c and g , which are chosen from the range of $[2^{-4}, 2^4]$. The results are presented in Table II, indicating that both classifiers can achieve satisfactory performance, and SVM slightly outperforms KNN. Nonetheless, given the decrease in prediction efficiency of KNN as the training data size increases, we decide to use the SVM classifier for subsequent experiments.

Influence of Speaking Volume and Pace: We divide *Trace-A* into two parts to investigate the impact of speaking volume and pace. Specifically, we utilize data from different devices, consisting of microphone and corresponding gyroscope signals, where the microphone data is collected at normal volume but with different paces, and at normal pace but with different volumes, to test the performance of GyroTalk. Experimental results are depicted in Figs. 14 and 15, which indicate that when

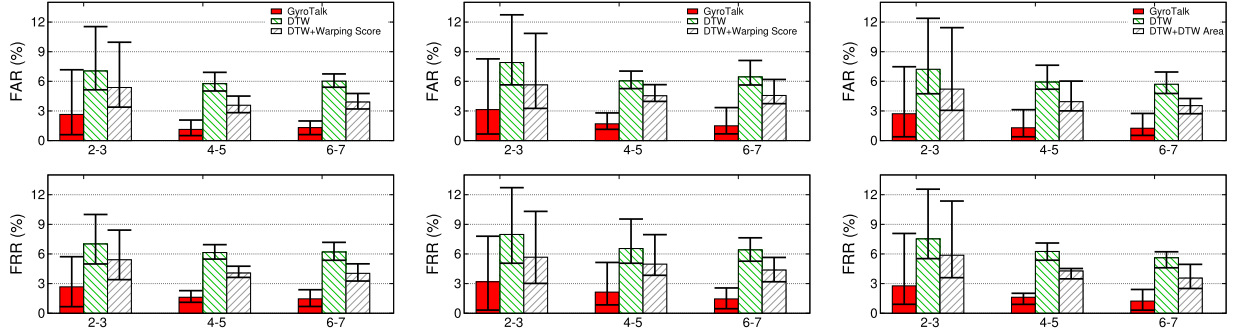


Fig. 17. FAR and FRR versus Length of voice command (left: watch, center: glasses, right: phone).

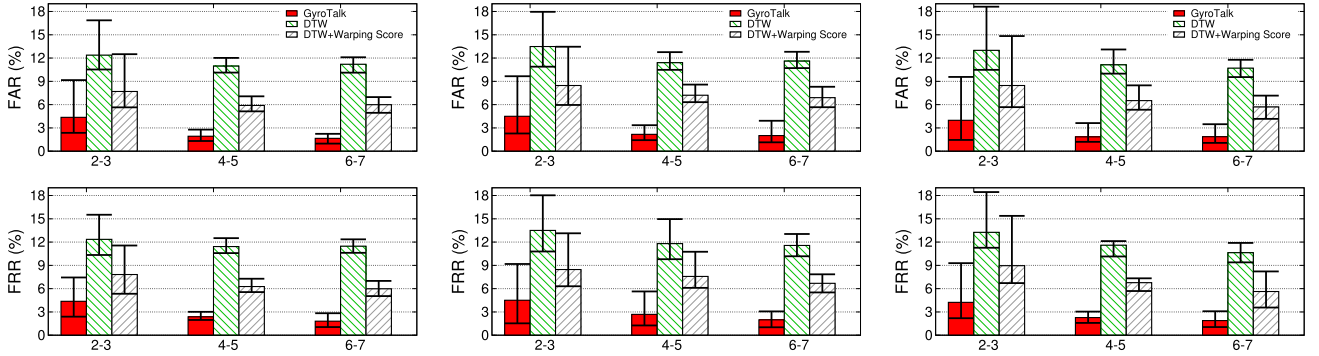


Fig. 18. FAR and FRR under spoofing attacks without priori knowledge(left: watch, center: glasses, right: phone).

speaking at a fast pace or low volume, the associated FARs and FRRs are approximately 3%-4%, which is 1%-2% higher than in other situations. Additionally, our findings suggest that the FRRs are more easily influenced by changes in speaking pace and volume compared to the FARs.

Influence of Voice Command Length: Fig. 17 demonstrates that, as the length of voice commands increases, both the FAR and FRR decrease. Specifically, the findings of the experiments show that when the length of voice command rises from 2 – 3 words to 4 – 5 words, the average FAR and FRR of GyroTalk on a smartphone decrease from 2.82% and 2.88% to 1.42% and 1.86%, respectively. These findings support our conjecture that longer voice commands provide stronger protection and decrease false rejections. Notably, GyroTalk outperforms the comparative techniques in all cases, e.g., for 6-7 words command, both types of errors for GyroTalk are only around 1.3%, whereas those of DTW with and without warping score still exceed 6% and 3%, respectively. Moreover, longer voice command (e.g., 10-11 words) can further improve GyroTalk’s performance. However, excessively lengthy voice commands for the voice assistant’s wake-up words will inevitably reduce user convenience. Typically, the optimal length of a voice wake-up command should contain 2-3 words.

B. Behavioral Variability

For behavioral biometric systems, behavioral variability is a significant problem. In this experiment, we examine the influence of long-term behavioral unpredictability on GyroTalk. To this end, we use *Trace-C* (collected six months after *Trace-A*) to

test GyroTalk trained and configured using *Trace-A*. As shown in Fig. 16, the FARs and FRRs do not vary extremely greatly over time in all cases. For example, for 2-3 word commands, we observe FARs are slightly increased from 3% (see Fig. 17) to around 3.5%, and for 4-7 word commands, both FARs and FRRs are no more than 2%, which means GyroTalk is resistant to long-term behavioral variability.

C. Combating Spoofing Attacks

We use *Trace-B* to analyze the security of GyroTalk based on the threat model discussed in Section III-A.

1) Spoofing Attack Without Prior Knowledge: Fig. 18 demonstrates the resilience of GyroTalk to attacks without prior knowledge. In comparison to normal usage scenarios, the FARs and FRRs of GyroTalk experience only slight increases (around 1%) for commands consisting of 2-5 words, and display nearly no change for commands consisting of 6-7 words. In contrast, both DTWs, with and without warping score, exhibit significant errors in all scenarios. The results indicate that, when subjected to attacks without prior knowledge, the performance of traditional voice recognition methods deteriorates significantly. This is likely due to the fact that such methods rely solely on the spectral characteristics of the user’s voice, making them more vulnerable to attacks that introduce subtle changes to the acoustic properties of the voice signal. Overall, these findings highlight the effectiveness of GyroTalk to voice recognition, which combines both audio and gyroscopic sensor data to improve accuracy and mitigate the impact of spoofing attacks. The results suggest that the security and dependability

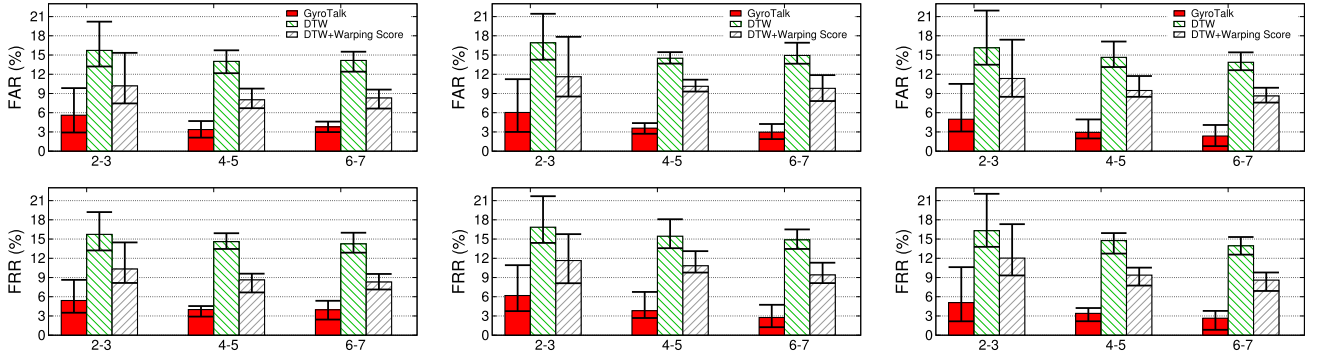


Fig. 19. FAR and FRR under spoofing attacks with priori knowledge(left: watch, center: glasses, right: phone).

TABLE III
MEASURED PROCESSING DELAY OF GYROTalk

Command Length	Processing Latency(s)	Time Breakdown (%)		
		Pre-processing	SA-DTW	Classification
7	1.357	0.75	99.22	0.03
5	0.882	0.93	99.03	0.04
3	0.521	1.22	98.71	0.07

of voice-based authentication systems can be improved by incorporating additional sensor data, particularly in scenarios where attacks without priori knowledge are a concern.

2) *Spoofing Attack With Prior Knowledge*: The results presented in Fig. 19 indicate that GyroTalk demonstrates stable performance even when subjected to increasingly complex spoofing attacks. Specifically, for commands consisting of 2-3 words, the error rate of GyroTalk increases by no more than 6%, while for commands consisting of 4-7 words, the increase is limited to no more than 3%. Furthermore, it is worth noting that in all circumstances, GyroTalk performs better than the other two studied approaches when it comes to resilience against spoofing assaults. More specifically, considering commands of 4-7 words, GyroTalk experiences an average FAR increase of approximately 1%, which is significantly lower than the corresponding FAR increases of over 7% and 5% observed for the other two methods. These findings suggest that GyroTalk is effective in mitigating the impact of various types of spoofing attacks, allowing it to maintain elevated accuracy rates even in the face of sophisticated attempts to deceive the system. Overall, the results demonstrate GyroTalk's superior robustness compared to alternative approaches and highlight its potential as a reliable solution for secure voice-based authentication.

D. GyroTalk Implementation

We implement GyroTalk as an app on a MOTO 360 Android Wear watch. Upon voice authentication by the user, the app operates surreptitiously in the background to capture gyroscope and microphone data, which are subsequently processed. Once the processing is finished, the app outputs a decision (accept or reject). We employ Fast DTW [35] to reduce the computational load from (n^2) to (n) while maintaining system performance. Table III displays the average processing delays measured for

GyroTalk app concerning voice commands of 7, 5, and 3 words, which are 1.357, 0.882 and 0.521 seconds, correspondingly. The lightweight nature of the app makes it feasible for deployment on low-end wearables.

VI. DISCUSSION AND LIMITATION

GyroTalk provides a novel approach to defend against replay attacks on advanced voice interfaces. It demonstrates robust defense without the need of additional hardware for wearable devices. However, GyroTalk remains some limitations. For instance, presently, it is only functional when the speaker remains still, otherwise, the FRR may be high. We speculate that this is an effect caused by the difference in sensor performance on different devices. Hence, this study identifies a future direction for research to eliminate the impact of the speaker's motion state by employing Independent Component Analysis (ICA) [36]. ICA can separate mixed signals that are simultaneously input and reconstruct them into signals based on independent components. In this case, the voice signal and the user motion noise signal are mixed based on independent components, so they can be separated by ICA. However, ICA also has the problem of how to select the independent components, so further research and improvement are needed in practice. Besides, GyroTalk is strongly dependent on the built-in gyroscope. When the wearable device cannot move properly (i.e., gyroscope movement is limited), the system cannot obtain effective gyroscope data, thus, the performance of GyroTalk will be greatly influenced. Finally, in order to create a mature solution, we will assess GyroTalk on a variety of wearables across a sizable user base.

VII. CONCLUSION

In this study, we create a voice authentication spoofing detection system called GyroTalk that uses the embedded gyroscope of wearables to record SMSs from speakers. The system captures the SMSs of speakers without the need for any additional hardware or cumbersome operations. The practicality of GyroTalk makes it applicable to wearables that can be worn in different positions. Extensive experiments are conducted, and the outcomes illustrate GyroTalk's robustness and effectiveness. Hence, this study identifies a future direction for research to eliminate the impact of the speaker's motion state by employing ICA. In addition, in order to develop a mature solution, we will

evaluate GyroTalk across a wide range of wearable products and a large user base.

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